

DBMS PROJECT PROPOSAL

Examination Scheduling & Result Processing System

Automating academic assessments through relational database design

THE CONTEXT

Scaling Academic Infrastructure

Modern universities manage thousands of concurrent enrollments.

- High data volume.
- Strict compliance requirements.
- Zero tolerance for errors.

Current System Limitations



Scheduling Conflicts

Overlapping student course enrollments.



Data Fragmentation

Isolated spreadsheets causing data inconsistencies.



Processing Delays

Manual transcript generation bottlenecks.

Target Outcomes



Automated Venue Allocation

Algorithmic room assignment preventing capacity overflow.



Data Integrity Enforcement

Strict relational constraints ensuring consistent records.



Real-Time Result Compilation

Instant metric aggregation and grading output.

Two Core Modules

01 Exam Scheduler

Timetable Generation: Conflict-free slot mapping.

Resource Logistics: Room and seat assignment.

Invigilator Duty: Faculty schedule balancing.

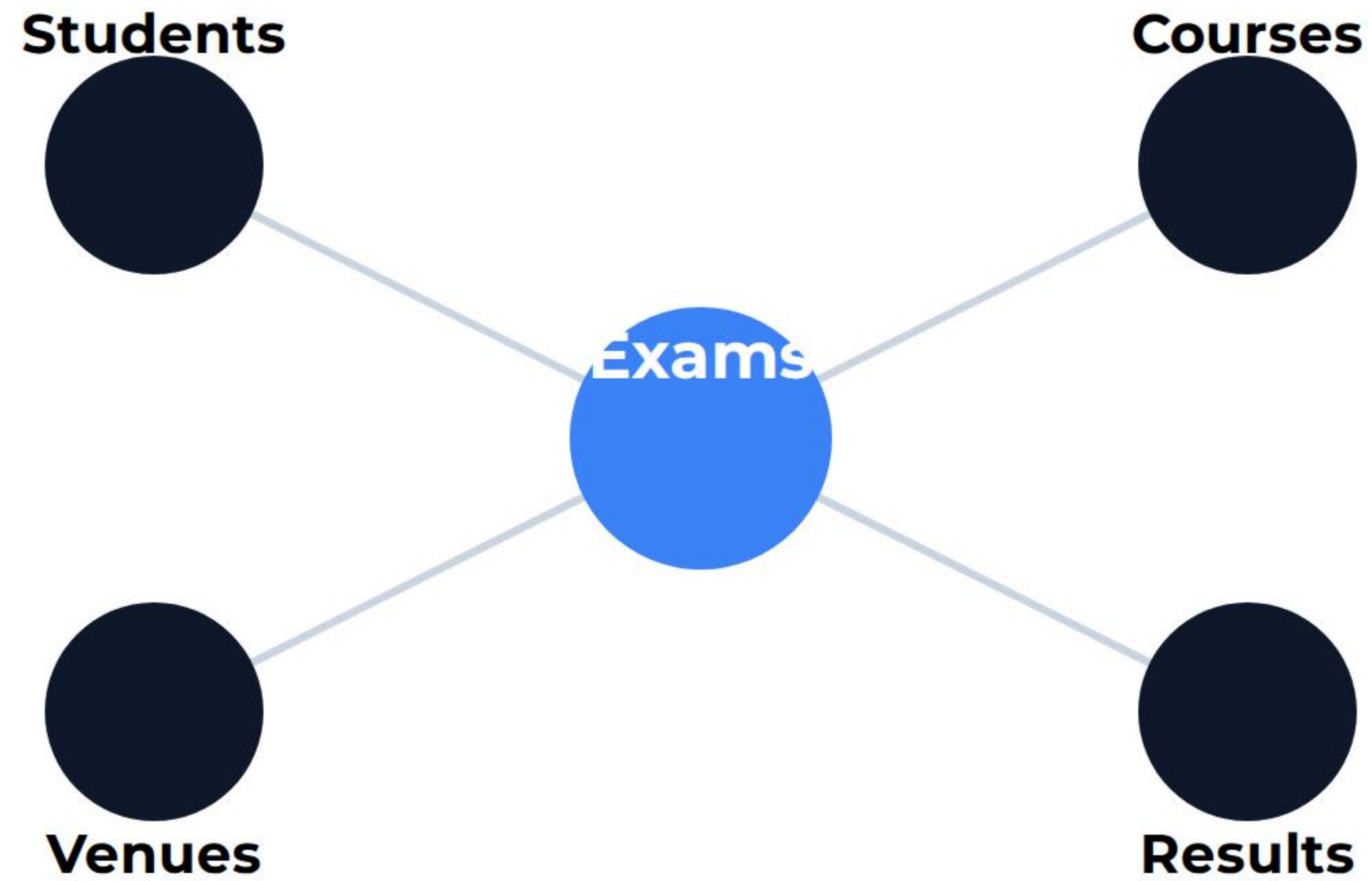
02 Result Processor

Score Capture: Secure grade entry portals.

Policy Calculation: Automated GPA computation.

Reporting: Standardized transcript generation.

Primary Data Entities



Proposed Technology Stack



Database

Relational SQL engine (MySQL / PostgreSQL).



Backend

Application logic and query processing APIs.



Frontend

Interactive interface for admins and students.

Immediate Next Steps



Phase 1

Proposal & Approval



Phase 2

ER Diagram Design



Phase 3

Normalization & Schema



Phase 4

SQL Implementation

Thank You

Open for feedback and questions.

Image Sources



https://studentportal.gu.se/sites/default/files/styles/100_10_4_xmedium_1x/public/2023-02/2656x1770-tentamen-R2C.jpg?h=cc598381&itok=xvblITa_

Source: studentportal.gu.se